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RESEARCH-ARTICLE

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Immersive Role-Playing: An Experience Report on a Promising Approach to Learning Requirements Elicitation

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Abstract

Developing the necessary skills in Software Engineering students to conduct interviews to elicit requirements is not a trivial activity. Immersive role-playing is a promising learning technique, allowing students to practice realistic interviews, receive real-time feedback, and improve their performance under pressure. This approach combines traditional role-playing with an immersive learning environment, offering an engaging and authentic experience that prepares students for the industry's challenges. This article aims to present an experience report on using immersive role-playing to teach the interview technique for eliciting software requirements. This experience was carried out with fifteen undergraduate students in Requirements Engineering. The results show the effectiveness of immersive role-playing in requirements elicitation, promoting increased engagement in tasks and authentic learning through feedback and reflection on mistakes made. The proposed immersive experience proved promising for improving education, engagement, and technical skills. Other skills relevant to training students in Software Engineering could also be worked on, such as oral communication, adaptability, and active listening.

CCS Concepts

• **Social and professional topics**; • **Professional topics**; • **Computing education**; • **Computing education programs**; • **Software Engineering education**;

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Keywords

Role-Playing, Immersive role-playing, Requirements elicitation, Immersive learning environment, Requirements engineering teaching

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1 Introduction

ACM's The growing demand for qualified professionals in Software Engineering, combined with rapid technological evolution, constantly challenges educational institutions to update their educational approaches. The article by Cavalcante et al. [1] highlights the importance of complementing theoretical teaching with practical experiences to better prepare students for the job market. This approach is important for developing the skills needed to work in the Information Technology (IT) field, where adapting to new methodologies, tools, and technologies is essential for success.

The teaching of Software Engineering faces significant obstacles, especially when providing students with practical experience. Often, these experiences are only acquired in professional internships, which can challenge all students to achieve the knowledge or live the experience equally. This gap between the theory taught in the classroom and the practice experienced in the job market highlights the need for innovative approaches to teaching Software Engineering, intending to prepare future professionals for the real challenges they will face. Among the essential skills for Software Engineers is the ability to apply Requirements Engineering (RE) techniques, especially those related to requirements elicitation.

The interview technique for requirements elicitation is one of the most effective tools professionals use to interact with clients and stakeholders [2] and the most widely used [4]. The ability

to elicit precise and comprehensive requirements from different stakeholders is closely linked to the interviewer's experience and interpersonal skills [5]. However, imparting this interviewing skill to undergraduate students becomes a challenge due to the diversity of factors that influence the success of this activity [6]. According to Bano et al. [4], students' main difficulties when simulating interviews are question formulation, omission, and communication.

Among the teaching strategies for developing interview skills in undergraduate students is the simulation of real situations. Several researchers have proposed methods and tools to explore these real-life situations [4, 7–10]. Some consider engaging students one of the biggest challenges in the classroom, and they face many difficulties in designing practical activities that can motivate them [11–13].

Immersive learning has emerged as a promising approach to Software Engineering education. Despite the diversity of definitions [14], immersive learning can be implemented using physical or digital media, methods, and technologies [15], facilitating its implementation in universities. Among physical and analog immersive methods, role-playing stands out. Role-playing is a technique based on the philosophy of learning by doing [16, 29], which has been used successfully in teaching requirements engineering [7, 17, 18]. It generates motivation and engagement, promoting different learning processes [19, 20].

Daun, Grubb, and Tenbergen [21] conducted a systematic literature review (SLR) and highlighted three main trends in instructional approaches to Requirements Engineering education, including role-playing and stakeholder involvement. This approach is effective in generating greater motivation and enthusiasm in students. Immersive learning environments emerge as an additional tool to engage students further and provide a more authentic learning experience besides role-playing.

An immersive learning environment is an educational environment that uses technologies such as Virtual Reality, Augmented Reality, and Mixed Reality for highly interactive and engaging educational experiences [30–32]. These environments give students the feeling of being in a virtual environment that simulates the real world, allowing them to explore, interact, and learn more effectively and engagingly [15]. Immersive learning environments enable students to experience real-world situations in a safe and controlled manner. They are applicable in various contexts, such as lectures, virtual laboratories, professional training, and simulations.

Immersive role-playing takes place when the role-playing is conducted in an immersive environment. According to Sakuma et al. [22], the construction of scenarios and tools allows the students to:

- Practice realistic interviews: realistic virtual environments enable students to familiarize themselves with different requirements elicitation situations;
- Receive feedback in real-time: immersive interaction with virtual interviewers makes it possible to identify strengths and weaknesses in the conduct of the interview, improving communication and information gathering;
- Improve their performance under pressure: the high-pressure interview simulation helps to identify flaws in planning, missing questions and improve communication under challenging situations.

This article aims to present an experience report on using immersive role-playing to teach the interview technique for eliciting software requirements. The experience was conducted in an immersive environment at PUCPR's Extended Reality Center. The experience report aims to answer the following research questions:

- RQ1: How do students perceive using an immersive environment to learn software requirements elicitation?
- RQ2: How do students perceive their learning using an immersive environment?
- RQ3: What are the challenges of using an immersive environment to teach software requirements elicitation?

The article is organized as follows: Section 2 presents related work. Section 3 describes the planning of the immersive experience. Section 4 details the execution of the interview simulation to elicit requirements. Section 5 presents the results. Finally, Section 6 concludes the work.

2 RELATED WORKS

In 2023, Daun, Grubb, and Tenbergen [26] carried out a systematic literature review (SLR), highlighting the importance of involving real or realistic stakeholders in teaching and learning [26]. They emphasized that authentic learning experiences can enhance students' Requirements Engineering (RE) knowledge and improve interpersonal collaboration and teamwork skills. To achieve this goal, role-playing has emerged as a trend in education, allowing the student to play a role in low-risk project-based collaborative instruction scenarios.

Several authors have explored the use of role-playing in the teaching of RE, especially in simulations. Andersson and Andersson [25] conducted studies in which students interacted with professional software engineers in realistic industrial scenarios. The results indicated that role-playing simulation promoted the learning of professional skills and generated a positive attitude among students toward the technique [25].

Ouhbi [13] applied role-playing in an RE course, in which students simulated situations as clients and developers, specifying and requesting functionalities. The process included three stages: tutorial, which combined traditional lectures with exercises based on discussions and requests arising from role-playing between the students and their project clients; project, in which each group of students carried out a project assignment in the role of construction managers, assigned by a professional client; social, involving conscious and formal assessment, as well as documentation of the personal development and progression of the student's social skills in their respective groups. The research demonstrated the effectiveness of role-playing in teaching RE in short periods and increasing student involvement [23].

Despite the predominance of scenario simulations in role-playing, the research by [18] and [24] aims to provide feedback so that students learn from their mistakes. The authors developed a software agent system model to monitor student behavior during role-playing training and provide real-time feedback. The model was effective in improving requirements development and student performance.

Following this line of learning from mistakes, Ferrari, Bano, and Zowghi [7] proposed a pedagogical approach for teaching

RE interviews that combines role-playing, peer review, and self-assessment. The aim is to promote experiential learning by allowing students to conduct mock interviews and then encourage reflection on their mistakes and those of their peers. They observed that this approach significantly reduced the amount of mistakes students made during the interviews.

Bano et al. [4] identified students' main errors in the initial learning phase when interviewing for requirements elicitation. These errors were mapped and categorized as follows: question formulation, omission of questions, order of interview questions, communication skills, analyst behavior, interaction with the client, teamwork, and planning. Among the most frequently observed errors were asking vague questions, ending the interview incorrectly, and not building rapport with the client.

In this work, we used immersive role-playing to develop technical and behavioral skills in the RE interview practice, providing students with an authentic learning experience similar to the challenges found in the industry.

3 PLANNING THE EXPERIENCE

We selected a Requirements Engineering 120-hour course offered in the 2nd semester of the bachelor's degree in Software Engineering at PUCPR. The Software Engineering curriculum at PUCPR is organized around four main competencies a student has to develop during a four-year bachelor's course. The Requirements Engineering course is compulsory and aims to teach students the complete process of Requirements Engineering. The course has six theoretical-practical classes per week using active learning methodologies. Elicitation techniques, such as the interview technique, are related to the competency: *Design computational solutions for diverse scenarios, combining methods and techniques appropriate to the context in a precise, critical, and innovative way.* The Learning Outcome (LO) that contributes to developing such competency is *identifying requirements using the most appropriate requirements elicitation technique for the problem and the context.*

The first step in planning the experience was establishing a scenario in which the students had no previous experience, so they had to develop their ability to elicit requirements. The scenario chosen was the development of the income tax system that the Federal Government uses for citizens to report their incomes and taxes by the end of each calendar year.

The second step was to create five personas to allow the students to experience different real-world situations, as shown in Figures 1-5. Creating personas is essential in providing students with an authentic learning experience, according to Mims [27]. When using personas, it is possible to simulate the different clients with whom they could interact. The personas were developed and adapted according to the recommendations of Preece, Roger, and Sharp [28]. They consisted of a name, a picture, the education profile, their income, their tax situation, their interests, and the challenges they face when filing a tax return. With the personas, it is possible to simulate realism, train empathy and communication, expose students to diverse scenarios, and prepare them to face situations similar to the job market [27].

The third step of the planning phase involved creating interview answer scripts, incorporating various unexpected situations that might occur in a natural interview environment. These included interruptions due to phone calls, clients giving evasive answers (yes or no), changes in clients' needs, and client indecision. The main goal was to prepare students to handle everyday challenges when eliciting requirements and develop strategies for overcoming these challenges.

Young Professional: Lucas

23 years old

- **Profession:** Junior Data Analyst
- **Education:** Recently graduated in Computer Science
- **Income:** Primarily from a full-time job as a data analyst, with some occasional freelance work in web development.
- **Tax Situation:** First-time taxpayer, with questions about how to include freelance income and the importance of educational deductions.
- **Personal Interests:** Technology, travel, and starting to invest in the stock market with small amounts.
- **Challenges:** Balancing professional life with continuing education in a postgraduate course.

Figure 1: Persona 1 – Young Professional

Retired Woman: Maria

68 years old

- **Profession:** Retired, former teacher
- **Education:** Degree in Education
- **Income:** Mainly from her retirement pension, supplemented by income from conservative investments.
- **Tax Situation:** Files taxes annually, with special attention to senior benefits and exempt income.
- **Personal Interests:** Gardening, reading, and volunteering with educational organizations.
- **Challenges:** Managing her finances to maintain a comfortable lifestyle, including health expenses.

Figure 2: - Persona 2 – Retired Lady

Middle-Aged Businesswoman: Ana

45 years old

- **Profession:** Owner of a clothing boutique
- **Education:** Degree in Fashion Design
- **Income:** Derived from the profit of her boutique, as well as rental income from properties.
- **Tax Situation:** Experienced in tax filing, focused on optimizing business-related deductions and investments.
- **Personal Interests:** Fashion, international business travel, and art.
- **Challenges:** Expanding the business in a competitive market and efficiently managing multiple income sources.

Figure 3: - Persona 3 – Middle Age Woman

Figure 4: - Persona 4 – Self-employed Woman

Figure 5: - Persona 5 – Self-employed Man

In the fourth phase, the group of students to whom the immersive role-playing experience would be applied was defined. It was decided that the group would consist of 15 students divided into five teams.

In the fifth phase, the stages for carrying out the experience were defined: students learning about the income tax system for which the requirements elicitation would take place, students drawing up a script to conduct the interview, students interviewing under the supervision of a professor, the professor giving feedback on the conduct of the interview and the professor applying an instrument for students to collect the students' perceptions of the experience. Three professors participated in the experience, referred to as T1, T2, and T3. Table 1 shows the questions designed to collect the students' perceptions of the experience.

4 EXECUTING THE EXPERIENCE

The experience occurred in three different rooms. Room 1, supervised by professor T1, received all the students and organized them into groups to wait for the immersive activity. The other two rooms (Room 2 and Room 3) are divided by a wall and connected by wired internet. Room 2 (Figure 2) had a TV, a desktop, a webcam, and a microphone for the students to interact with a digital avatar representing the persona. In Room 2, professor T2 acted as an observer, and at the end of each interview, T2 gave immediate feedback to the students. Room 3 (Figure 3) had a screen, a desktop, an iPhone, a cell phone holder, a headset, and a microphone for the professor to represent the client (persona) and interact with the students. Professor T3 acted as the interviewee and was presented to the student as a digital avatar. The execution is detailed below:

- **General instructions**: professor T2 explained to the students the aim of the activity, which was to simulate a natural user interview environment. T2 presented the system context in which the students would have to conduct the interviews and the rules for conducting the activity.
- **Interview script**: drawing the script is crucial for the activity's success. To do this, the students had 90 minutes to research the context of the system to be developed and draw up the interview script under the supervision of professors T1 and T2. The students could choose the strategy for drawing up the scripts and were allowed to ask questions.
- **Interview**: professor T1 stayed in Room 1 with all the students, while T2 went to Room 2 and received each group separately, as shown in Figure 6. The students interacted with a TV using a microphone and viewing an avatar representing the client. Professor T2 took notes during the interview to provide individualized feedback at the end of the activity. Professor T3 (Figure 7) was in the interviewee's room (Room 3) and was responsible for selecting the persona and avatar representing the client being interviewed. T3 interacted with the interviewers (students) through a microphone and viewed them through a camera. The voice changed according to the avatar chosen. A different persona and corresponding avatar were selected for each team.
- **Feedback**: professor T2 provided individualized feedback to each team after each interview, highlighting appropriate and inappropriate actions during the requirements elicitation interview. The team's reactions to the feedback varied. Some were passive, accepting all the professor's observations, while others sought to deepen their learning by asking questions.

Table 1: - Questions applied at the end of the immersive experience

RQ	Objective	Question
RQ1	To evaluate the perception of the immersive environment experience	How did you feel carrying out this activity?
RQ2	To evaluate the perception of learning	What did you learn from this activity?
RQ3	To evaluate the perception of the challenges faced	What were the main challenges you faced in this activity?

Figure 6: - Room 2 – Students interviewing the client

Figure 7: - Room 3 – Professor T3 representing the client

- **Data collection**: an online form with the questions presented in Table 1 was available for each student to answer individually.

5 RESULTS

This section presents the general observations made by professors T2 and T3 about the student's behavior during the interviews with each team and the results obtained from the data collection, organized according to the research questions.

5.1 Observations on students behavior

Below is a breakdown of the observations on students' behavior, separated by team, indicating appropriate behavior (Table 2) and inappropriate behavior (Table 3) during the interview. Each behavior has an identifier: PB for proper behavior, followed by a number, and IB for inappropriate behavior, followed by a number. We put an ST (student team) identifier to identify the teams, followed by a number.

Concerning appropriate behaviors (PB), shown in Table 2, teams ST1, ST2, ST3, and ST4 had some common behaviors, such as explaining the purpose of the interview to the client (PB01), asking broad questions followed by specific ones (PB02), not being imposing (PB03) and trying to get to know the client's profile (PB04). The ST2 team was more prepared because they showed important behaviors for understanding the requirements, such as active listening (PB05), seeking a deeper understanding of the business (PB06), and improvising by asking questions outside the script based on the client's doubts (PB07). Team ST5 had an imposing attitude and faced more difficulties than the others.

Regarding inappropriate behavior (IB), Table 3, the following errors were common to most teams: following the script and not responding to the unexpected (IB3), not listening actively (IB4), not asking direct questions (IB5), asking technical questions instead of business questions (IB6) and a lack of planning and coordination between team members (IB12). The ST2 and ST5 teams recorded the interview without asking for the clients' permission (IB07), which is unacceptable behavior in any context. The ST4 and ST5 teams were impatient with the client's questions (IB9). They did not respond well to simulations of the client's lack of knowledge.

The ST5 team did not explain the purpose of the interview (IB10) and took too long to start the interview (IB11), which could be considered disrespectful to the client´s time.

5.2 RQ1 - How do students perceive using an immersive environment to learn software requirements elicitation?

Analysis of the results revealed a variety of student's perceptions of the experience in the immersive environment. Some participants expressed feelings of fear and initial strangeness, as evidenced by the comments: ***"Afraid of the 3D doll used to represent the interviewee"*** **and** ***"Strange, it was something new, and I didn't know what to expect".*** However, these emotions were overcome as the students got involved in the activity.

The participants highlighted positive aspects of the experience, such as the opportunity to put into practice the content covered in class and the feeling of significant learning. Comments reflect this perception, such as: "*The dynamics were very interesting, so we could put into practice the content covered in class"* and "*It was a very important experience to understand how a real interview works*".

In addition, the students described the experience as innovative, enjoyable, and productive. There were reports of enthusiasm at the novelty of the experience, as evidenced by the comments: ***"I felt fascinated, since it was a completely new and different experience"*** and ***"A very different and productive experience. I really enjoyed it".***

The students' comments indicate that immersive environments can effectively foster meaningful learning and engage students. Although some students initially experienced discomfort, the majority responded positively to the experience. They highlighted its

Table 2: - Students' appropriate behaviors (PB)

Proper Behavior	ST1	ST2	ST3	ST4	ST5
PB1 - Explain the purpose of the interview to the interviewee	X	X	X	X	
PB2 - Start with broad questions and then ask specific questions about the business	X	X			
PB3 - Not having an imposing attitude	X	X	X	X	
PB4 - Getting to know the client's profile		X	X	X	
PB5 - Active listening		X			
PB6 - Seeking a deeper understanding of the business		X			
PB7 - Not sticking to the script and knowing how to improvise based on the client's questions		X			

Table 3: - Students' inappropriate behaviors (IB)

Behavior	ST1	ST2	ST3	ST4	ST5
IB1 – Not asking for the interviewee's name	X				
IB2 - Not asking questions to understand the interviewee's profile	X				X
IB3 - Following the script and not responding to the unexpected	X		X	X	X
IB4 - Not listening actively	X		X	X	X
IB5 - Not asking direct questions	X		X	X	X
IB6 - Asking technical questions	X		X	X	X
IB7 - Not asking permission to record		X			X
IB8 - Asking irrelevant questions			X	X	X
IB9 - Showing impatience with client questions				X	X
IB10 - Not explaining the purpose of the interview					X
IB11 - Taking too long to start the interview					X
IB12 - Lack of planning and coordination between team members	X		X	X	X

benefits, such as the practical application of knowledge acquired in lectures and case studies and the opportunity to encounter similar situations in the real world.

The immersive experience proved to be interesting for authentic learning. The students were able to put into practice the knowledge they had acquired in the class, improving their skills in:

- **Requirements elicitation:** the need to formulate effective questions and adapt to the reality of the virtual interviewee contributed to the achievement of the LO - Identify requirements using the RE technique best suited to the problem and context.
- **Interpersonal communication:** interaction with the avatar requires students to have skills such as clear communication, empathy, and adaptability, preparing them to deal with different types of interviewees in real situations.
- **Teamwork:** collaboration in building the interview script and conducting the activity promoted the development of interpersonal skills such as teamwork, conflict resolution, and joint decision-making.

Using immersive environments to teach requirements elicitation was well received by the students. The experience provided an opportunity to develop practical skills, deepen theoretical understanding, and promote greater student engagement and interest in the subject. However, further research is needed to explore the long-term effects of this approach and identify effective strategies to maximize its benefits in the educational context.

5.3 RQ2 - How do students perceive their learning using an immersive environment?

Table 4 shows some quotes from the students' perceptions of their learning. They reported the importance of the practical experience of interviewing a software product user and the opportunity to learn from their mistakes.

Participants highlighted the importance of adapting the language and interview style to the interviewee's knowledge, as evidenced by the comments of S9: ***"To conduct the interview, it was necessary to identify the interviewee's knowledge of the subject and adapt the questions accordingly."*** And also by S12: ***"The ideal is to adapt to the client's language, speaking in a totally informal way, if necessary, but at the same time, having a great knowledge of the content being covered is mandatory."***

In addition, the participants reported having learned how to structure the questions in order to extract meaningful requirements from the answers and the importance of delving deeper into the subjects covered during the interview. Comments such as S13 reflect this perception: ***"I learned that I should go deeper into the subject and always ask the interviewee"***.

Another significant contribution of the immersive environment was the opportunity to receive immediate and targeted feedback. Participants emphasized the importance of this feedback in correcting mistakes and improving their interviewing skills. As participant S15 said, ***"The professor's feedback was extremely valuable. It added much to the learning process, concretizing successes and pointing out mistakes."***

Table 4: - Students' perceptions about their learning

Std	Perception
S1	*"That when we interview, we **shouldn't go in thinking that the interviewee already has knowledge of the area**"*
S2	*"I learned in practice **how to conduct an interview in the right way**"*
S3	*"I've learned that I need to be more **objective with some questions and go more in-depth with** some of the interviewee's observations."*
S4	*"I learned that it's **important to get to know the interviewee and adapt the language** used in the interview (more technical, less technical)"*
S5	*"How to **prepare questions for a** recruitment **interview** and how to conduct an interview correctly?"*
S6	*"I learned how to **conduct** the interview **correctly**"*
S7	*"I learned to pay **attention to detail when conducting** a requirements elicitation interview. Always formulate a clear and organized script to collect all possible information from the user."*
S8	*"I was able to **learn a few things about income tax and how to ask the interviewee good questions**, and after the questions the professor helped me by showing me that we need to answer any questions in the interview itself so that there are no misunderstandings or unforeseen events after the interview."*
S9	*"To conduct the interview, it **was necessary to identify the interviewee's knowledge of the subject** and adapt the questions accordingly"*
S10	*"Best **analysis of an** interview **script**"*
S11	*"I've learned how to **put together questions** for an interview and that we should delve deeper into subjects and know about them in detail, as well as asking what I don't understand about the interviewee's speech."*
S12	*"The ideal is to **adapt to the client's language**, speaking in a totally informal way, if necessary, but at the same time, having a great knowledge of the content being covered is mandatory."*
S13	*"I've learned to **structure questions in such a way that it's possible to extract a requirement** from their answer. In addition, I've learned that I should go deeper into the subject and always ask the interviewee something I don't know or don't understand."*
S14	*"One thing I learned **was to always ask and question the interviewee about something I don't know.** At the end of the interview, the professor's feedback was very important"*
S15	*"In terms of learning, the biggest gain was certainly the **experience of having something real**, **having immediate feedback, the surprise of the answers, the interaction**. The professor's feedback was extremely valuable, it added a lot to the whole learning process, concretizing any successes and pointing out mistakes."*

The comments suggest that using an immersive environment can be an effective tool for promoting learning. By providing a more realistic and interactive experience, the immersive environment allows participants to develop practical interview skills, adapt to different contexts, and receive constructive feedback for continuous improvement.

5.4 RQ3 - What are the challenges of using an immersive environment to teach software requirements elicitation?

Regarding RQ3, which aimed to obtain the students' perception of the main challenges they faced during the experience (Table 5), the following were highlighted:

- **Preparing questions:** the need to formulate effective queries and adapt to the reality of the virtual interviewee requires students to have clear, concise, and targeted communication skills, revealing gaps in the construction of precise and in-depth questions.
- **Knowing the interviewee:** the diversity of profiles and knowledge of the avatars (personas) required the students to be flexible and adaptable, exposing the difficulty in identifying the level of understanding of the interviewee and adapting the questions according to the reality of the interview.
- **Time and adaptation:** the dynamics of the immersive interview, with limited time and a surprise virtual interviewee, demanded improvisational, adaptability, and time management skills from the students, highlighting the need to improve these skills to deal with unpredictable situations in actual interviews.

Regarding RQ3, which aimed to gather students' perceptions about the challenges of using the immersive environment, Table 5 presents the highlighted points.

6 DISCUSSION

6.1 Professors' Observations

The professor T2 observation during the experiment was valuable for providing feedback on the students' learning process. Although the intervention was short (180 minutes), observing the student's behavior concerning the successes and errors expected in the interview process conducted by beginners in requirements elicitation was possible.

The proposed experience generated more engagement on the part of the students, demonstrating that role-playing in requirements

Table 5: - Students' perceptions to the challenges during the activity.

Std	Perceptions
S1	*"Maintain **eye contact** with the interviewee."*
S2	*"The main challenges were in relation to the **interview questions**, our group asked superficial questions and we weren't able to extract all the necessary information."*
S3	*"The **creation of the interview script**, because we needed to know more about income tax for a more succinct interview."*
S4	*"Not knowing exactly what to expect in terms of **the level of understanding of the client interviewed on the subject.**"*
S5	*"The **development of questions**"*
S6	*No answer*
S7	*"The biggest challenge was to **structure a simple script** that contains all the items that need to be covered, but with room to adapt to any information the user may have"*
S8	*"I think that, in itself, it was getting to **know more about income tax, which I didn't know anything about and what it was**, so I tried to learn the basics so that I could answer any questions and have a clean and clear interview with the interviewee."*
S9	*"The **group's lack of knowledge on the subject made it** difficult to find good questions to ask the interviewee, who also didn't have much experience in the subject"*
S10	*"**Be more curious**"*
S11	*"**Difficulties in setting up the interview in such a way that it** feels like an informal conversation so that the interviewee is comfortable."*
S12	*"**Observe and adapt to the client's profile.**"*
S13	*"I had a hard time **constructing the interview script**, because I had to think and research in depth about a subject I wasn't familiar with."*
S14	*"The main challenge was with the **script; we thought** more about the system's questions than the person's personal characteristics, such as their occupation, age, etc."*
S15	*"**Maybe time**, being **able to collect everything in time to be able to put together a coherent interview**. Also, the question of being a surprise interviewee, not knowing the needs and having to adapt during the interview."*

elicitation learning provides a more motivating attitude. Being in a different environment from the traditional one, with immersive and playful technologies, exposes the students to an experience closer to what will be found in industry, awakening more motivation to carry out the tasks. These results confirm the research of Ouhbi [23], which shows that role-playing is an effective tool for teaching requirements elicitation in short periods and creates engagement on the part of the students.

Observing the students' mistakes was essential for the teaching-learning process. Based on the errors the students made, the professors were able to provide more precise and individualized feedback. The students could reflect on their errors, analyze what they could have done differently, and identify what skills they needed to develop, whether technical or behavioral. These findings corroborate the studies by [7], [18] and [24].

Still, on the subject of student inappropriate behavior observations, it was possible to classify them according to the work of Bano et al. [4], divided into seven classes:

- **1.Error with formulating questions**: no vague or imprecise, technical or irrelevant questions should be asked. Questions in this category were present in 80% of the teams. Vague or imprecise questions can lead to various interpretations on the interviewee's part, resulting in misunderstandings, for example, ***"What can be improved in the system?".*** This question does not direct the interviewee to discuss specific aspects of the system, such as tax calculations, user interface, data security, and regulatory compliance, which are relevant. Technical questions, such as ***"Would you like the system to have multi-factor authentication to add an extra layer of security?"*** may also not be appropriate, as it cannot be assumed that the person has detailed technical knowledge. These questions can intimidate the client and damage the relationship. Irrelevant questions, such as ***"Are you married or single?"*** do not contribute to the development of the system or do not align with the client's profile. Such questions take up time during the interview and result in irrelevant data being obtained, which can lead to incorrect or duplicate requirements being drawn up.
- **2.Omission of questions**: 20% of the teams omitted relevant questions. When essential questions are not asked, there is a risk of missing crucial information for the development of the system, which can lead to gaps in the requirements identified. For example, one could miss essential information if they do not ask questions to understand the interviewer's profile. This, in turn, can discourage the interviewee from asking critical questions, resulting in a system that does not effectively meet the users' needs. Ensuring that all relevant questions are asked is essential for a complete and accurate elicitation of requirements.

- **3.Order of questions**: in the interview, this refers to the sequence in which the questions are asked, from the beginning to the end. This order creates a flow of conversation that should guide the clients logically to understand the project's vision. Some students (40% of the teams) made mistakes when starting or ending the interview and arranging the questions incorrectly. For example, they began with direct questions before an overview of the project, such as***, "What would you like to see implemented in terms of system security?"*** instead of addressing specific client difficulties, such as ***"Do you face any difficulties when filling in your tax return?".*** Requirements engineers must develop a questioning strategy that considers the interview context, making it easier to elicit specific requirements.
- **4.Communication skills**: This is a challenge in teaching, in general, and becomes even more evident when teaching requirements elicitation. In this experience, 80% of the teams faced difficulties using the interviewees' answers to extract requirements or had other communication failures. For example, one interviewee began to share details about how he filed his total personal income tax return to obtain a larger refund. Still, the interviewers ignored this information and continued with questions that were not aligned with the context presented by the interviewee. The students showed significant difficulties in dealing with unexpected situations, which had a negative impact on communication between the client and the interviewer.
- **5.Interviewer behavior**: behavior during interviews can influence the attitude of clients and affect their responses. In 80% of the interviews, the interviewers did not introduce themselves or ask for the interviewee's name, going straight to the predefined questions. Some students showed immaturity when being corrected by the interviewee, while others laughed unprofessionally. These attitudes can cause discomfort for the client being interviewed, resulting in a lack of cooperation in answering essential questions during the requirements elicitation process.
- **6.Client interaction**: the success of an interview depends on the interaction between interviewer and client, and the interviewer must create a welcoming environment that promotes communication. In 40% of the interviews, the student interviewers failed to introduce themselves or even ask for the interviewee's name, going straight into a series of direct questions and ending the dialog abruptly without a final summary. Developing a contextualized questioning strategy is essential to obtain specific requirements and improve client-interviewer interaction. According to Bano et al. [4], this strategy involves establishing a relationship, understanding the client's business process and current problems, and summarizing the findings to confirm understanding.
- **7.Planning and teamwork**: in 20% of the interviews, the students had not planned a division of tasks for asking questions and taking notes, resulting in long pauses and early termination of the interviews. This reflected a lack of planning in defining the objectives of obtaining information, selecting the right questions, and organizing the interaction during the interview.

A significant finding of this study is that a considerable part of the students, approximately 60% of the teams, do not practice active listening during interviews. Active listeners demonstrate skills such as avoiding hasty judgments and providing appropriate and encouraging verbal and non-verbal feedback. This skill is crucial for effective communication between both parties during the interview, and its absence can result in misunderstandings and even stress for the interviewee, who feels ignored by the interviewer. The practice of active listening is neither trivial nor easy to teach, highlighting a gap that needs to be investigated to improve this skill among students and, consequently, their ability to deal with unexpected situations.

6.2 Students' Perceptions

The reports are positive when we analyze the students' perceptions of the immersive environment. According to their perception, the close-to-real-life experience was motivating and generated a better understanding of interview techniques for eliciting requirements. The immersive experience significantly contributed to developing technical professional skills, made the students understand their mistakes in practice, and allowed them to reflect on them. The students highlighted the importance of learning by doing and in a different environment.

Delivering feedback in real time was a fundamental practice during the intervention. Feedback was given as soon as each team finished its interview. This moment was critical for highlighting the successes and mistakes of each team, intending to stimulate critical reflection on their performance. The timing for this activity is crucial; once time passes, the experiences tend to fade into their memory. According to the students' perceptions, learning from mistakes was particularly valuable, as they considered that the most significant learning lessons came from the situations in which they made mistakes.

One point highlighted by the students was the need to have more time to prepare for the activity and to receive more details about the interviewees. Time is vital for this activity and completing it in less than 180 minutes is challenging. A viable alternative would be to extend the time of the activity, giving the students a more extended period to prepare. Thus, a 120-minute timeframe could be dedicated to preparing and researching the interview topic. Then, another 120 minutes could be devoted to carrying out the interview, professor feedback, and student reflection, making a total experience of 240 minutes.

7 CONCLUSIONS

Analysis of the student's perceptions of using an immersive environment in teaching requirements elicitation reveals the potential of this approach to improve student learning and engagement. By recognizing their essential skills and weaknesses, students highlight the immersive experience as a valuable tool, even though it is still little explored. This experience not only awakens curiosity but also improves essential skills, contributing to forming complete professionals prepared for future challenges in this area.

The experience highlighted a limitation in developing non-technical skills, especially oral communication. The students struggled to deal with unexpected situations and adapt to the written

script during the interviews. Although the professors provided feedback at the end of the activity, the lack of opportunity for the students to redo the activity and learn from their mistakes represents a point to be considered in improving this experience.

As prospects for future work, we intend to refine the experience to replicate it. Regarding the preparation step, we will give students more time to formulate appropriate questions. During the interview, we must better prepare the students to become more flexible in dealing with adverse situations and strengthen their oral communication skills. The post-interview aims to investigate how students deal with the data and information collected to optimize this learning process. These are opportunities to improve the teaching methodology and the student's requirements engineering skills, contributing to a more complete and practical education.

References

[1] Victória Cavalcante, Catarina Costa, Laura Sarkis, Daricélio Soares, and Saulo Freitas. 2023. Contributions of an Extension Course Focused on Good Software Engineering Practices for Students and IT Professionals. In *Anais do XXII Simpósio Brasileiro de Qualidade de Software*, novembro 07, 2023, Brasília/DF, Brasil. SBC, Porto Alegre, Brasil, 301–310.

[2] Alan Davis, Óscar Dieste Tubío, Ann Hickey, Natalia Juristo, and Ana Moreno. 2006. Effectiveness of Requirements Elicitation Techniques: Empirical Results Derived from a Systematic Review. In 14th IEEE International Conference on Requirements Engineering (RE). IEEE, 176-185. https://doi.org/10.1109/RE.2006.17.

[3] Alessio Ferrari, Paola Spoletini, and Sourav Debnath. 2022. How do requirements evolve during elicitation? An empirical study combining interviews and app store analysis. Requirements Engineering 27. https://doi.org/10.1007/s00766-022-00383-7.

[4] Muneera Bano, Didar Zowghi, Alessio Ferrari, Paola Spoletini and Beatrice Donati. 2019. Teaching requirements elicitation interviews: an empirical study of learning from mistakes. Requirements Engineering 24, 259-289. https://doi.org/10.1007/s00766-019-00313-0.

[5] Daniel Méndez Fernández, WagnerStefan, Marcos Kalinowski, *et al.* 2017. Naming the pain in requirements engineering. Empirical Software Engineering 22, 5, 2298-2338.

[6] PictureIrit Hadar, Pnina Soffer and Keren Kenzi. 2014. The role of domain knowledge in requirements elicitation via interviews: an exploratory study. Requirements Engineering 19, 2, 143-159. https://doi.org/10.1007/s00766-012-0163-2.

[7] Alessio Ferrari, Paola Spoletini, Muneera Bano, and Didar Zowghi. 2019. Learning Requirements Elicitation Interviews with Role-Playing, Self-Assessment, and Peer-Review. In 2019 IEEE 27th International Requirements Engineering Conference (RE). IEEE, Jeju, Korea (South), 28-39. https://doi.org/10.1109/RE.2019.00015.

[8] Binnur Görer and Fatma Aydemir. 2023. Emerging Technologies in Requirements Elicitation Interview Training: Robotic and Virtual Tutors.

[9] Alessio Ferrari, Paola Spoletini, M. Bano, and D. Zowghi. 2020. Role-playing and role reversal. Requirements Engineering 25, 4, 417-438.

[10] A Adrian Rusu and Robert Russell and Remo Cocco. 2011. Simulating the Software Engineering interview process using a decision-based serious computer game. In Proceedings of the 16th International Conference on Computer Games (CGAMES). IEEE, 235-239.

[11] Stephen John Quaye, Shaun R. Harper, and Sumun L. Pendakur. 2019. Student Engagement in Higher Education: Theoretical Perspectives and Practical Approaches for Diverse Populations. Routledge.

[12] Stephan Krusche, Nadine von Frankenberg, Lara Marie Reimer, and Bernd Bruegge. 2020. An interactive learning method to engage students in modeling. In Proceedings of the ACM/IEEE 42nd International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET '20). ACM, New York, NY, USA, 12-22. https://doi.org/10.1145/3377814.3381701.

[13] Sofia Ouhbi and Nuno Pombo. 2020. Software Engineering Education: Challenges and Perspectives. In 2020 IEEE Global Engineering Education Conference (EDUCON). IEEE, Porto, Portugal, 202-209. https://doi.org/10.1109/EDUCON45650.2020.9125353.

[14] Andreas Dengel. 2022. What Is Immersive Learning? In Proceedings of the 8th International Conference of the Immersive Learning Research Network (iLRN). Vienna, Austria, May 30-June 4.

[15] Stylianos Mystakidis and Vangelis Lympouridis. 2023. Immersive Learning. Encyclopedia 3, 396-405. https://doi.org/10.3390/encyclopedia3020026.

[16] John Dewey. 1986. Experience and education. The Educational Forum 50, 3, 241-252.

[17] Didar ZowghiSuresh ParyaniSuresh Paryani. 2003. Teaching requirements engineering through role playing: lessons learnt. In Proceedings of the 11th IEEE International Requirements Engineering Conference. IEEE, Monterey Bay, CA, USA, 233-241. https://doi.org/10.1109/ICRE.2003.1232754.

[18] Taichi Nakamura, Un Kai and Yuki Tachikawa. 2014. Requirements engineering education using expert system and role-play training. In 2014 IEEE International Conference on Teaching, Assessment and Learning for Engineering (TALE). IEEE, Wellington, New Zealand, 375-382. https://doi.org/10.1109/TALE.2014.7062566.

[19] Laura M. Castro. 2023. Role-playing software architecture styles. In 2023 IEEE 20th International Conference on Software Architecture Companion (ICSA-C). IEEE, L'Aquila, Italy, 171-174. https://doi.org/10.1109/ICSA-C57050.2023.00045.

[20] Claudia H. Montenegro, Hernán Astudillo and Maria Clara Gomez. 2017. ATAM-RPG: A role-playing game to teach architecture trade-off analysis method (ATAM). In 2017 XLIII Latin American Computer Conference (CLEI). IEEE, Cordoba, Argentina, 1-9. https://doi.org/10.1109/CLEI.2017.8226416.

[21] Marian Daun, Alicia M. Grubb and Bastian Tenbergen. 2021. A Survey of Instructional Approaches in the Requirements Engineering Education Literature. In 2021 IEEE 29th International Requirements Engineering Conference (RE). IEEE, Notre Dame, IN, USA, 257-268. https://doi.org/10.1109/RE51729.2021.00030.

[22] Hiroshi Sakuma, Hideyuki Takahashi, Kohei Ogawa, Hiroshi Ishiguro. 2023. Immersive role-playing with avatars leads to adoption of others' personalities. Frontiers in Virtual Reality 4, 1025526. https://doi.org/10.3389/frvir.2023.1025526.

[23] Sofia Ouhbi. 2019. Evaluating Role Playing Efficiency to Teach Requirements Engineering. In 2019 IEEE Global Engineering Education Conference (EDUCON). IEEE, Dubai, United Arab Emirates, 1007-1010. https://doi.org/10.1109/EDUCON.2019.8725045.

[24] Yuki Tachikawa and Taichi Nakamura. 2017. Education for requirements elicitation using group-work and role-play. In 2017 IEEE Global Engineering Education Conference (EDUCON). IEEE, Athens, Greece, 780-783. https://doi.org/10.1109/EDUCON.2017.7942935.

[25] Niclas Andersson and Pernille Andersson. 2010. Teaching professional engineering skills: Industry participation in realistic role-play simulation. Proceedings of the 6th International CDIO Conference (CDIO 2010).

[26] Marian Daun, Alicia M. Grubb, Viktoria Stenkova and Bastian Tenbergen. 2023. The Field of Requirements Engineering Education. In 2023 IEEE 35th International Conference on Software Engineering Education and Training (CSEE&T). IEEE, Tokyo, Japan, 119-119. https://doi.org/10.1109/CSEET58097.2023.00023.

[27] Clif Mims. 2003. Authentic Learning: A practical introduction and guide for implementation. 6.

[28] Jennifer Preece, Helen Sharp and Yvonne Rogers. 2015. Interaction Design: Beyond Human-Computer Interaction. Wiley, Hoboken, NJ. ISBN: 978-1-119-02075-2.

[29] Yuichiro Anzai and Herbert A. Simon. 1979. The theory of learning by doing. Psychological Review 86, 2, 124-140. https://doi.org/10.1037/0033-295X.86.2.124.

[30] [Ahmed Alnagrat, Rizalafande C. Ismail, and Syed I. S. Zulkarnain. 2022. A Review of Extended Reality (XR) Technologies in the Future of Human Education: Current Trend and Future Opportunity. Journal of Human Reproductive Sciences 1, 81-96. https://doi.org/10.11113/humentech.v1n2.27.

[31] Zuheir Khlaif, Allam Mousa, and Mageswaran A/L Sanmugam. 2024. Immersive Extended Reality (XR) Technology in Engineering Education: Opportunities and Challenges. Technology, Knowledge and Learning 1-24. https://doi.org/10.1007/s10758-023-09719-w.

[32] Filipe Fernandes, Diego Castro, Claudia Rodrigues, and Claudia Werner. 2022. Development of the Software Engineering Education Virtual Classroom Prototype: An Experience Report. WEI 85-96. https://doi.org/10.5753/wei.2022.222590.

[33] Conference Location:El Paso, Texas USA